

Lecture 10: The Chain Rule

GOAL: Be able to differentiate functions of the form $f(g(x))$.

Summary:

$\sin(x^2)$ e^{3x}

$$[f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\frac{d}{dx}[f(g(x))] = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$[g(x)^p]' = p[g(x)]^{p-1} \cdot g'(x) \quad [b^x]' = b^x \cdot \ln b$$

Theorem (Chain Rule) Let f and g be diff fns. Then

$$\text{Format I: } \frac{d}{dx}(f \circ g) = \frac{df}{dg} \cdot \frac{dg}{dx}$$

$$\text{Format II: } [f(g(x))]' = f'(g(x)) \cdot g'(x)$$

$$\text{Ex 1 } \frac{d}{dx}(x^{\frac{3}{2}}) = \frac{3}{2} x^{\frac{1}{2}} \quad x^{\frac{3}{2}} = [x^{\frac{1}{2}}]^3$$

$$x \mapsto x^{\frac{1}{2}} \mapsto (x^{\frac{1}{2}})^3 = f(g(x))$$

$$g(x) = x^{\frac{1}{2}}$$

$$f(g) = g^3$$

$$\begin{aligned} \text{Format I: } \frac{df}{dx} &= \frac{df}{dg} \cdot \frac{dg}{dx} = \frac{d}{dg}(g^3) \cdot \frac{d}{dx}(x^{\frac{1}{2}}) \\ &= 3g^2 \cdot \frac{1}{2} x^{-\frac{1}{2}} = 3(x^{\frac{1}{2}})^2 \cdot \frac{1}{2} x^{-\frac{1}{2}} \\ &= \frac{3}{2} x^{\frac{1}{2}} \end{aligned}$$

$$\text{Format II: } [(x^{\frac{1}{2}})^3]' = f'(g) \cdot g'$$

$$\begin{aligned}
 &= \underbrace{[g^3]'}_{\text{w.r.t. } g} \cdot \underbrace{[x^{\frac{1}{2}}]'}_{\text{w.r.t. } x} \\
 &= 3g^2 \cdot \frac{1}{2} x^{-\frac{1}{2}} = \dots = \frac{3}{2} x^{\frac{1}{2}}
 \end{aligned}$$

NOTE: This gives us a way to show the Power Rule holds if the exponent is rational.

$$x^{\frac{m}{n}} = (x^{\frac{1}{n}})^m$$

Integers

Ex2 $[(\sin x)^2]'$ | $g(x) = \sin x$
 $f(g) = g^2$

Format I: $\frac{df}{dx} = \frac{df}{dg} \cdot \frac{dg}{dx} = \frac{d}{dg}(g^2) \cdot \frac{d}{dx}(\sin x)$
 $= 2g \cos x = 2 \sin x \cos x$

Format II: $[(\sin x)^2]' = f'(g(x)) \cdot g'(x)$
 $= \underbrace{[g^2]'}_{\text{w.r.t. } g} \cdot \underbrace{[\sin x]'}_{\text{w.r.t. } x}$

$= 2g \cos x = 2 \sin x \cos x$
 $x \mapsto x^2 + x + 1 \rightarrow e^{(x^2 + x + 1)}$

Ex3 $y = e^{x^2 + x + 1}$

$g(x) = x^2 + x + 1$

$f(g) = e^g$

$$\frac{dy}{dx} = \frac{d}{dg}(e^g) \cdot \frac{d}{dx}(x^2+x+1) = e^g \cdot (2x+1) \\ = (2x+1)e^{x^2+x+1}$$

NOTE: It is very common to skip the step of writing f as a fn of g

$$\left[e^{x^2+x+1} \right]' = \left[e^{x^2+x+1} \right]' \cdot \underbrace{\left[x^2+x+1 \right]'}_{\text{w.r.t. } x} \\ = e^{x^2+x+1} (2x+1)$$

Ex 4 $w(x) = \tan(\cos(x^2))$
 $x \mapsto x^2 \mapsto \cos(x^2) \mapsto \tan(\text{---})$

$$w'(x) = \sec^2(\cos(x^2)) \cdot [\cos(x^2)]' \\ = \sec^2(\cos(x^2)) \cdot -\sin(x^2) \cdot \underbrace{[x^2]'}_{2x} \\ = -2x \sin(x^2) \sec^2(\cos(x^2))$$

Applications of The Chain Rule

Ex 5 Diff $y = (x^3-1)^{100}$ $\begin{cases} g(x) = x^3-1 \\ f(\text{---}) = (\text{---})^{100} \end{cases}$

$$y' = 100(x^3-1)^{99} \cdot [x^3-1]' = 100(x^3-1)^{99} (3x^2)$$

$$= 300x^2(x^3-1)^{99}$$

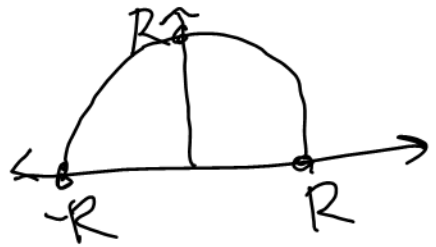
Theorem (Generalized Power Rule)

Let g be a diff fcn and p a real number. Then

$$[(g(x))^p]' = p(g(x))^{p-1} \cdot g'(x)$$

If $g(x) = x$, this is the usual power rule.

Ex 6 Let $R > 0$. The semicircle of radius R is given by $y = \sqrt{R^2 - x^2}$



Find y'

$$y' = [\sqrt{R^2 - x^2}]' = [(R^2 - x^2)^{\frac{1}{2}}]'$$

$$= \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} \cdot [R^2 - x^2]'$$

$$= \frac{1}{2} (R^2 - x^2)^{-\frac{1}{2}} \cdot (-2x) = \dots = \frac{-x}{\sqrt{R^2 - x^2}}$$

Ex 7 What's $\frac{d}{dx}(2^x)$?

$$[2^x]' = [e^{\ln(2^x)}]' = [e^{x(\ln 2)}]' \quad \left| \begin{array}{l} g(x) = (\ln 2)x \\ f(-) = e^{(-)} \end{array} \right.$$

$$= e^{x(\ln 2)} \cdot \ln 2$$

$$= 2^x \cdot \ln 2$$

In general, if $b > 0$,

$$\frac{d}{dx} (b^x) = b^x \cdot \ln(b)$$

Ex 8 (Damped Pendulum) A pendulum's position is measured from its angle from its resting point.

$$S(t) = \underbrace{[e^{-t} \cos t]}_{\text{rad}}$$

 Find $v(t)$

$$\begin{aligned} v(t) = S'(t) &= [e^{-t}]' \cos t + e^{-t} [\cos t]' \\ &= -e^{-t} \cos t + e^{-t} (-\sin t) \\ &= [-e^{-t} (\cos t + \sin t)] \frac{\text{rad}}{\text{s}} \end{aligned}$$

Ex 9 The radius of a circle is evolving over time, how fast is the area changing as a fn of time.



$$\begin{aligned} A &= \pi r^2 \\ A(t) &= \pi [r(t)]^2 \\ \frac{dA}{dt} &= \pi 2[r(t)] \frac{dr}{dt} \end{aligned}$$

Ex 10 π radians $= 180^\circ \Rightarrow \frac{\frac{\pi}{180} \text{ rad}}{1^\circ}$

$$\sin(x^\circ) = \sin\left(\underbrace{\frac{\pi}{180} x}_{\text{radians}}\right)$$

$$\begin{aligned} [\sin(x^\circ)]' &= \left[\sin\left(\frac{\pi}{180} x\right)\right]' = \cos\left(\frac{\pi}{180} x\right) \cdot \frac{\pi}{180} \\ &= \frac{\pi}{180} \cos(x^\circ) \end{aligned}$$